



Seminar on Economic Policy and the Welfare State

***The 2018 Parallel Society Legislation and the
Immigrant-Native Employment & Unemployment Gap***

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June 3, 2026

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ABSTRACT

This paper examines whether Denmark’s 2018 Parallel Society legislation causally narrowed the native–non-Western immigrant employment and unemployment gaps across Danish municipalities. Using a two-way fixed effects difference-in-differences design applied to administrative panel data from Statistics Denmark covering all 98 municipalities over 2010–2022, I compare gap dynamics in the 15 designated municipalities against a restricted cohort of 24 structurally comparable control municipalities. The preferred specification finds that the legislation is associated with a gradual and statistically significant narrowing of the employment gap of approximately 1.70–1.75 % points by 2022, which is supported by Callaway–Sant’Anna doubly robust estimator at –1.96 % points. Effects emerge with a two-year implementation lag and are concentrated among newly designated municipalities, consistent with abrupt network disruption as the operative mechanism. The unemployment gap cannot be given a causal interpretation due to persistent pre-trend violations. These findings suggest that while the legislation achieved its stated goal of reducing the number of designated areas, it also generated measurable short-run labour market disruption for affected non-Western immigrant communities.

1 Introduction

Denmark is one of Europe’s most expansive social housing sectors, with approximately one sixth of the population residing in non-profit housing estates (Landsbyggefonden, 2025)[4]. Within this sector, immigrants and their descendants now constitute 31.1% of all social housing residents, reflecting both their relative economic position and historical municipal placement policies that directed newly arrived refugees toward available social housing (Damm and Rosholm, 2005)[3]. In response, successive Danish governments pursued increasingly coercive spatial interventions over three decades, culminating in the 2018 Et Danmark Uden Parallelsamfund legislation (The 2018 Parallel Society legislation) which is the most extensive version of Danish ghetto policy. This policy has introduced binding population composition caps, mandatory demolition and estate sale programmes, and compulsory dispersal of non-Western immigrant households from designated areas (Regeringen, 2018).[10]. Scholars argue that the 2018 legislation used urban policy to classify and regulate populations. By making the 50 % non-Western resident threshold the decisive criterion for ghetto designation, the policy linked social disorder to ethnic composition and used housing interventions to reshape where targeted populations could live (Lundsteen, 2023)[5]. Sørensen, and Christensen (2024)[11] further document that the legislation’s development plans collectively reduced non-profit family housing units by over 6,300 across 17 designated areas, raising substantial concerns about displacement, resegregation, and the loss of social capital that underpins labour market participation.

The OECD (2026) reports a significant reduction in the number of designated areas between 2018 and 2024, yet notes that persistent employment disadvantage continues to characterize the remaining areas. This raises the central research question of this paper:

Did the 2018 Parallel Society legislation causally affect the native–non-Western immigrant employment and unemployment gaps in designated Danish municipalities?

To answer this question, the paper employs a difference-in-differences design applied to municipality-year panel data from Statistics Denmark administrative registers [12] over 2010–2022, comparing employment and unemployment gap dynamics in the 15 designated municipalities against a restricted cohort of structurally comparable control municipalities. The analysis contributes to the quasi-experimental municipality-level evidence on the employment effects of the 2018 legislation.

2 Background and Motivation

Awareness of concentrated disadvantaged residents in some social housing estates led Danish governments to introduce several policy response. In 2004, it published its first strategy against ghettoisation (Regeringen, 2004)[8], and in 2010 it introduced the more ambitious ”Returning the Ghetto to Society” plan, which created a formal list of vulnerable housing areas ranked by immigrant share, unemployment, crime, education, and income (Regeringen, 2010)[9]. The 2018 parallel societies legislation went furthest, setting binding targets by 2030 through stricter rules on resident composition, demolition in some areas, and wider use of flexible letting. Nielsen, Nordberg and Andersen (2023)[6] argues that this legislation changed the role of social housing in affected municipalities. Instead of mainly providing housing to people with limited options, the sector increasingly became a tool for managing who lives in particular estates. The OECD (2026)[7] reports that the number of vulnerable and parallel society areas fell considerably between 2018 and 2024, confirming the policy’s effectiveness in reducing ghetto areas, but notes

that persistent employment disadvantage continues to characterize the remaining areas. This persistence motivates the central empirical question of this paper: whether the legislative framework that produced this physical transformation simultaneously affected the labor market integration of the non-Western immigrant residents it displaced.

3 Data Source, Panel construction and Descriptive Statistics

3.1 Data Sources

The data set support a quasi-experimental evaluation of the 2018 Danish Parallel Society (Parallelsamfund) legislation’s effect on the native vs Non-Western (NW)¹ immigrant employment and unemployment gap across Danish municipalities. All data are drawn from Statistics Denmark [12] administrative registers. Data access and panel construction follow a fully programmatic workflow to ensure transparency and replicability. All five administrative registers — RAS200, RAS206, FOLK1C, BOL, and INDKP were retrieved directly from the Statistics Denmark (DST) StatBank via its API. API queries were constructed and executed in Python 3.13, with the requests library managing HTTP calls to the DST StatBank endpoints and the pandas library handling data parsing, variable construction, and panel merging. The complete data construction pipeline is available upon request and can be fully reproduced from the raw API outputs.

3.2 Panel construction

3.2.1 Dataset 1: Municipality-Country-Year Panel

The second dataset is a three-dimensional panel structured at the municipality $\times$ origin country $\times$ year level. It covers the same 98 municipalities and 13 years as Dataset 1, disaggregated across 20 non-Western origin countries, yielding 25,480 municipality-country-year observations (98 x 20 x 13). This dataset is the intermediate product from which the Bartik shift-share instrument in Dataset 1 is derived, and it supports all origin-group-disaggregated analyses.

The 20 origin countries tracked in Dataset 2 represent the principal non-Western immigrant communities resident in Denmark. Syria ranks fourth with 190,698 cumulative resident-years, a figure that grows substantially after 2015 with a year on year growth of 97.3 in 2015 after the refugee crisis as shown in figure 1 that forms an important pre-treatment contextual event in this study.

3.2.2 Dataset 2: Municipality-Year Panel

The main panel construction covering all 98 Danish municipalities over 13 consecutive years (2010-2022), yielding 1,274 municipality-year observations with zero missing values across all 17 variables. The panel spans the pre-treatment period (2010-2017) and the post-treatment period (2018-2022), with 2017 serving as the reference year immediately preceding the Parallel Society legislation. Fifteen municipalities are coded as treated (TREAT = 1) based on [13] 2018 ghetto list by Transport, Bygninger og Boligministeriet (2018); the remaining 83 municipalities serve as the control group.

¹Non-Western (NW) immigrants are defined following Statistics Denmark’s classification as persons born outside Denmark whose parents were also born outside Denmark and originate from a country outside the EU/EEA, North America, Australia, and New Zealand. The 20 specific origin countries tracked in this panel are: Afghanistan, Bosnia-Herzegovina, China, India, Iran, Iraq, Lebanon, Morocco, Nepal, Pakistan, Philippines, Russia, Somalia, Sri Lanka, Syria, Thailand, Turkey, Ukraine, Vietnam, and Yugoslavia

The primary labour market data are drawn from RAS200, which classifies all individuals aged 16-64 as employed, unemployed, or outside the labour force as of week 48 each year. The employment and unemployment rates of native and non western immigrants are constructed using

$$\text{NATIVE_EMP_RATE}_{mt} = \frac{\text{Employed native Danes aged 16–64 in municipality } m \text{ at time } t}{\text{Working-age native population aged 16–64 in municipality } m \text{ at time } t} \times 100$$

$$\text{NW_EMP_RATE}_{mt} = \frac{\text{Employed NW immigrants aged 16–64 in municipality } m \text{ at time } t}{\text{Working-age NW immigrant population aged 16–64 in municipality } m \text{ at time } t} \times 100$$

The difference between native and non-Western immigrant rates within each municipality-year cell are calculated as:

$$\text{EMP_GAP}_{mt} = \text{NATIVE_EMP_RATE}_{mt} - \text{NW_EMP_RATE}_{mt}$$

$$\text{UNEMP_GAP}_{mt} = \text{NATIVE_UNEMP}_{mt} - \text{NW_UNEMP}_{mt}$$

A positive EMP_GAP indicates that natives have a higher employment rate than NW immigrants in that municipality-year. A negative UNEMP_GAP indicates that NW immigrants face higher unemployment rates than natives.

FOLK1C population counts, measured on January 1st of each year, are merged to provide the total non-Western immigrant stock $\text{TOTAL_NW_STOCK}_{mt}$ and the non-Western immigrant share of the working-age population $\text{NW_IMMIG_SHARE}_{mt}$ for each municipality-year cell.

Dataset 1 (municipality-year) is constructed by collapsing Dataset 2 across origin countries to produce the Bartik instrument, then merging the resulting municipality-year AVG_EMP_RATE_NAT values with municipality-year outcomes from RAS200 and controls from BOL and INDKP. For each municipality, year, and origin country, the stock of non-Western immigrants of that origin from FOLK1C is combined with the national employment rate of that origin group from RAS206 to produce a stock-weighted average predicted employment rate:

$$\text{AVG_EMP_RATE_NAT}_{mt} = \frac{\sum_c (\text{STOCK}_{cmt} \times \text{EMP_RATE_NAT}_{ct})}{\sum_c \text{STOCK}_{cmt}}$$

where the summation runs over all 20 origin countries c present in municipality m at time t presented in dataset 2. This variable captures what the non-Western employment rate in a municipality would be if immigrants performed exactly at their national origin-group average, net of any local municipal effect. Critically, $\text{AVG_EMP_RATE_NAT}_{mt}$ is identical between treated and control municipalities, with a value of 54.50 in both groups and $p = 1.00$. This confirms that the two groups share similar non-Western immigrant origin compositions on average, strengthening the validity of the treated-control comparison by ruling out compositional confounding.

The social housing share (SH_SHARE) from the BOL housing register and the natural logarithm of average disposable income (LOG_INCOME) from INDKP are merged at the municipality-year level which are the time varying controls.

3.2.3 Descriptive Statistics

Table 1 presents mean comparisons of key variables between the 15 treated and 83 control municipalities, pooled across all 13 years. Treated municipalities exhibit systematically larger employment gaps (24.40 pp versus 21.28 pp, $p < 0.001$), higher social housing shares (22.38% versus 18.49%, $p < 0.001$), larger NW immigrant shares (7.81% versus 6.25%, $p < 0.001$), and substantially larger NW immigrant stocks (6,760 versus 1,363 persons on average, $p < 0.001$). This means treated municipalities are, by construction, those with elevated social housing and NW immigrant concentrations. Importantly, the Bartik shift-share instrument (AVG_EMP_RATE_NAT) is identical between treated and control groups (54.50% in both, difference = 0.00), confirming that the two groups share the same NW immigrant origin-country composition on average.

Table 1: Descriptive Statistics

Variable	All Mean	Treated Mean	Control Mean	Std. Dev.
Employment gap (native – NW immigrant, pp)	21.755	24.401	21.277	6.678
Unemployment gap (native – NW immigrant, pp)	-3.405	-3.103	-3.459	3.085
Native employment rate (%)	76.215	75.624	76.322	4.091
NW immigrant employment rate (%)	54.461	51.223	55.046	8.181
NW immigrant emp. rate, stock-weighted (%)	54.018	52.367	54.316	5.101
Native unemployment rate (%)	2.498	2.581	2.483	0.917
NW immigrant unemployment rate (%)	5.902	5.683	5.942	3.227
Municipal unemployment rate (%) [total]	2.824	2.947	2.802	0.937
Social housing share of total stock (%)	19.084	22.383	18.488	12.127
Log average disposable income	12.327	12.299	12.332	0.170
NW immigrant share of working-age pop. (%)	6.489	7.813	6.249	4.090
NW immigrant stock, all ages (count)	2,188.990	6,760.200	1,362.867	4,312.051
AVG_EMP_RATE_NAT (%)	54.500	54.500	54.500	0.00
Number of municipalities	98.000	15.000	83.000	
Observation years	2010–2022			
Municipality-year observations	1,274.000	195.000	1,079.000	

Note: Significance of treated-control differences is based on Welch two-sample t -tests with unequal variances: *** $p < 0.001$, ** $p < 0.01$, and * $p < 0.05$. The AVG_EMP_RATE_NAT difference of 0.00 confirms identical baseline observable characteristics (Social housing shar, income, NW share) between treated and control groups, validating the Bartik instrument construction.

A descriptive trajectory of the native–non-Western (NW) immigrant employment gap across the full 2010–2022 panel in fig 2 where two principal episodes define the time series: the Syrian Refugee Crisis of 2015–2016, the COVID-19 pandemic shock of 2020. This has been identified from the visualization of the absolute employment rates and the gap level (Panel A), the gap disaggregated by treatment and control municipality groups (Panel B), and year-on-year changes in the gap (Panel C). First, the Syrian refugee crisis of 2015 produced the largest single-year gap widening in the panel (+2.76 pp), driven not by labour market deterioration among established NW immigrants but by the labour force reclassification of newly arrived refugees, as evidenced by stable NW unemployment rates alongside falling NW employment rates and the near-complete convergence of treated and control municipality gaps under quota-based dispersal of the refugees across all municipalities. Second, the COVID-19 pandemic of 2020 counter-intuitively narrowed the gap by 1.10 pp, reflecting the differential sectoral exposure of the two groups:

NW immigrants' over-represented in essential-sector jobs (cleaning, logistics, elder care) that never closed, while natives are heavier in hospitality and retail that shut down due to covid.

4 Empirical Strategy

4.1 Identification Framework

The central empirical challenge is to isolate the causal effect of the 2018 Parallel Society legislation on the native–non-Western immigrant employment and unemployment gaps from confounding factors that differ systematically between treated and control municipalities. This study employs a two-way fixed effects difference-in-differences (TWFE DiD) design, compares how the employment and unemployment gap changed after 2018 in treated municipalities versus control municipalities, after removing all time-invariant municipality characteristics and all year-specific shocks.

The DiD design rests on the parallel trends assumption: in the absence of the 2018 legislation, the employment gap in treated municipalities would have evolved along the same trajectory as in control municipalities. This assumption is not directly testable, but the event study specification provides an empirical test by estimating whether treated and control municipalities exhibited differential pre-treatment trends during 2010–2016.

A key threat to DiD validity is the inclusion of structurally dissimilar control municipalities. The full 83-municipality control group includes small, rural areas with negligible social housing and NW immigrant populations fundamentally unlike the large urban treated municipalities. A restricted cohort is therefore constructed, excluding controls where 2010 baseline social housing or NW immigrant shares fall at or below five percent, retaining 24 comparable controls alongside all 15 treated municipalities (39 municipalities; 507 observations). This binding constraint not only mitigates the risk control group not serving as a valid counterfactual for the designated treated municipalities but also makes the parallel trend assumption validated and thereby remove any pre-trends from the treated and control groups.

4.2 Baseline Specification

The analysis is conducted on a balanced municipality-year panel comprising $M = 98$ municipalities observed across $T = 13$ years from 2010 to 2022, yielding $N = 1,274$ municipality-year observations in the full sample and $N = 507$ in the restricted sample. The panel structure allows for the inclusion of two-way fixed effects that absorb all time-invariant municipality characteristics, such as geographic location, historical settlement patterns, and local institutional infrastructure, as well as common macroeconomic shocks affecting all municipalities in a given year.

The baseline two-way fixed effects DiD specification is:

$$\text{Gap}_{mt} = \alpha_m + \alpha_t + \beta (\text{Treated}_m \times \text{Post}_t) + \gamma' \mathbf{X}_{mt} + \varepsilon_{mt}. \quad (1)$$

where Gap_{mt} is either EMP_GAP_{mt} or UNEMP_GAP_{mt} for municipality m in year t ; α_m is a municipality fixed effect absorbing all time-invariant unobserved heterogeneity; α_t is a year fixed effect capturing aggregate shocks common to all municipalities; Treated_m is a time-invariant binary indicator for designated municipalities; and Post_t is a binary indicator equal to one for $t \geq 2018$. The coefficient of interest is β , which identifies the average treatment effect on the treated (ATT) which is the change in the employment gap in treated municipalities relative to control municipalities after 2018, net of municipality

and year fixed effects.

The vector $\mathbf{X}_{mt}$ includes three time-varying municipality-level controls: social housing share; log average disposable income and non-Western immigrant share of the working-age population. These controls absorb variation in municipal economic conditions and demographic composition that may correlate with both treatment assignment and gap dynamics. All standard errors are clustered at the municipality level to account for arbitrary serial correlation within municipalities over the 13-year panel, following Bertrand, Duflo, and Mullainathan (2004)[1].

4.3 Event Study Specification

To test the parallel trends assumption and characterise the dynamic trajectory of treatment effects, the baseline specification is extended to an event study DiD:

$$\text{Gap}_{mt} = \alpha_m + \alpha_t + \sum_{\tau \neq 2017} \delta_\tau (\text{Treated}_m \times \mathbb{1}[t = \tau]) + \gamma' \mathbf{X}_{mt} + \varepsilon_{mt}. \quad (2)$$

where the summation runs over all years τ in the panel except the reference year $\tau = 2017$, which is omitted to normalise all coefficients relative to the year immediately preceding the legislation. The interaction terms $\text{Treated}_m \times \mathbb{1}[t = \tau]$ are created for each calendar year, with $\text{Treated}_m = 1$ for designated municipalities and $\mathbb{1}[t = \tau] = 1$ when the observation year equals τ . The coefficients δ_τ are therefore interpretable as the average gap difference between treated and control municipalities in year τ relative to the year 2017, after removing municipality and year fixed effects.

The year 2017 is selected as the reference year for three reasons. First, it is the last pre-legislation year, providing the cleanest pre-law baseline for normalisation. Second, it post-dates the peak of the Syrian refugee crisis from 2015 to 2016, avoiding contamination of the reference period by the refugee shock. Third, choosing the year immediately before treatment is standard practice in the event study literature, as it minimises the time-distance between the baseline and the treatment onset.

Pre-treatment coefficients δ_τ for $\tau \in \{2010, 2011, 2012, 2013, 2014, 2015, 2016\}$ provide the empirical test of parallel trends: if the identifying assumption holds, these coefficients should be statistically indistinguishable from zero and exhibit no systematic trend. Post-treatment coefficients δ_τ for $\tau \in \{2018, 2019, 2022\}$ capture the dynamic treatment effect, allowing assessment of whether the legislation's impact materialises immediately or builds over time.

4.4 COVID-19 Control Specification

The COVID-19 pandemic introduced a major macroeconomic shock in 2020–2021 that may have affected employment outcomes differentially across treated and control municipalities. Treated municipalities, concentrated among larger urban centres, tend to have higher shares of non-Western immigrants employed in contact-intensive service sectors such as hospitality, cleaning, retail, and transport, which were disproportionately exposed to pandemic-induced demand contractions and lockdown measures. Failure to account for this differential shock could conflate the pandemic effect with the legislation effect for years 2020 and 2021. To address this concern, a COVID-controlled variant of both specifications is estimated. A COVID interaction term is added to the baseline model:

$$\text{Gap}_{mt} = \alpha_m + \alpha_t + \beta (\text{Treated}_m \times \text{Post}_t) + \phi (\text{Treated}_m \times \text{COVID}_t) + \gamma' \mathbf{X}_{mt} + \varepsilon_{mt}. \quad (3)$$

where COVID_t is a binary indicator equal to one for $t \in \{2020, 2021\}$. The coefficient ϕ captures any differential COVID-19 shock to the gap in treated municipalities during the pandemic years, while β continues to identify the legislation effect net of this shock.

In the event study specification, including both $\text{Treated}_m \times \text{COVID}_t$ and the individual year interactions $\text{Treated}_m \times \mathbb{1}[t = 2020]$ and $\text{Treated}_m \times \mathbb{1}[t = 2021]$ would generate perfect collinearity, since the former is a linear combination of the latter two. The COVID-controlled event study therefore replaces the 2020 and 2021 individual year interactions with the single $\text{Treated}_m \times \text{COVID}_t$ term:

$$\text{Gap}_{mt} = \alpha_m + \alpha_t + \sum_{\tau \notin \{2017, 2020, 2021\}} \delta_\tau (\text{Treated}_m \times \mathbb{1}[t = \tau]) + \phi (\text{Treated}_m \times \text{COVID}_t) + \gamma' \mathbf{X}_{mt} + \varepsilon_{mt}. \quad (4)$$

Under this formulation, the δ_τ coefficients for $\tau \in \{2010, \dots, 2016, 2018, 2019, 2022\}$ capture the pure legislation effect in each year, while ϕ absorbs the average differential COVID shock across 2020–2021.

4.5 Heterogeneous Treatment Effects by Treatment History

The 15 treated municipalities differ in their historical exposure to parallel society designation. Eleven municipalities were designated under both the 2010 and 2018 criteria, referred to as continuously treated; four municipalities appear only on the 2018 list, referred to as newly treated; and six municipalities that appeared on the 2010 list are absent from the 2018 list, referred to as exited. To assess whether the post-2018 effect varies by treatment history, the baseline specification is extended to a heterogeneous treatment effects model:

$$\text{Gap}_{mt} = \alpha_m + \alpha_t + \beta_1 (\text{New}_m \times \text{Post}_t) + \beta_2 (\text{Cont}_m \times \text{Post}_t) + \beta_3 (\text{Exit}_m \times \text{Post}_t) + \gamma' \mathbf{X}_{mt} + \varepsilon_{mt}. \quad (5)$$

where New_m , Cont_m , and Exit_m are time-invariant binary indicators for the newly treated, continuously treated, and exited groups respectively. Table 4 provides the full composition of each group where the municipality classification is based on the 2010 government report *Ghettoen tilbage til samfundet: Et opgør med parallelsamfund i Danmark*. [9] This regression splits treatment history to see whether the 2018 legislation affected different types of municipalities differently. This specification serves three purposes. First, β_1 and β_2 allow us to compare post-2018 effect that differ between newly and continuously designated municipalities, capturing heterogeneous exposure intensity. Second, the Exit group coefficient β_3 constitutes a falsification test: municipalities that exited designation status should not exhibit a post-2018 treatment effect if the legislation is the operative mechanism, since they were no longer subject to its obligations after 2018. Thus β_3 should be statistically insignificant and close to zero, as these municipalities were no longer subject to designation obligations after 2018, $\text{Post}_t = 1$ for $t \geq 2018$. Third, comparing β_1 and β_2 allows assessment of continuously treated municipalities, having faced sustained policy pressure, may exhibit stronger or more persistent effects than newly designated ones.

4.6 Robustness check: Callaway–Sant’Anna Heterogeneity-Robust Estimator

After running the regression we find that the employment gap outcome satisfies the parallel trend but the unemployment gap outcome exhibits a mild but persistent pre-trend within the restricted control cohort. It may account for the possibility that municipalities with high non-Western unemployment before the

reform may have continued to experience different unemployment trends for structural reasons unrelated to the 2018 policy.

A rigorous remedial strategy implements the Callaway and Sant’Anna (2021)[2] doubly robust difference-in-differences estimator, which relaxes the parallel trends assumption by conditioning on observable municipality characteristics and using only never-treated municipalities as the comparison group for each treatment cohort. Since all 15 treated municipalities share a common first-treatment year of 2018, a single-cohort design, the group-time average treatment effect $ATT(g = 2018, t)$ is estimated for each calendar year t as:

$$ATT(g = 2018, t) = \mathbb{E} [Y_t(1) - Y_t(0) \mid G = 2018]. \quad (6)$$

where the counterfactual outcome $Y_t(0)$ is identified via outcome regression adjustment using the observable control vector (social housing share, income, NW share). Standard errors for the Callaway–Sant’Anna estimator are computed via cluster bootstrap with 500 iterations, with clustering at the municipality level to maintain consistency with the baseline TWFE specification.. The Callaway–Sant’Anna framework offers two advantages over the standard TWFE event study in this context. First, by conditioning on observable characteristics, it reduces the confounding contribution of structural differences in baseline unemployment rates that drive the pre-trend in the unadjusted comparison. Second, its pre-treatment $ATT(g = 2018, t)$ estimates for $t < 2018$ constitute a more demanding test of conditional parallel trends than the unadjusted event study, since the regression adjustment explicitly accounts for the observable dimensions along which treated and control municipalities differ.

5 Empirical Results

This section presents the empirical results across four specifications: the baseline TWFE DiD on the full 98-municipality sample, the preferred restricted control cohort, the COVID-controlled augmented specification, and the heterogeneous treatment effects model and the robustness check results using Callaway–Sant’Anna Heterogeneity-Robust Estimator. The primary outcome of interest is the employment gap (EMP_GAP_{mt}); the unemployment gap ($UNEMP_GAP_{mt}$) is reported as a secondary outcome with the caveat that its parallel trends assumption is not satisfied.

5.1 Baseline TWFE DiD: Full Sample

The baseline TWFE DiD estimates for the employment gap across the full 98-municipality sample, we find that the legislation coefficient is positive and statistically significant: $\beta = +1.117$ ($SE = 0.557, p = 0.045$) without controls and $\beta = +1.231$ ($SE = 0.607, p = 0.042$) with controls. This indicates that the employment gap widened by approximately 1.1–1.2 percentage points in treated municipalities relative to control municipalities after 2018. However, the event study specification reveals that the full-sample estimates are compromised by pre-trend violation four of seven pre-treatment years exhibit significant deviations from zero. This pattern indicates that treated and control municipalities in the full sample were already diverging before the 2018 legislation.

5.2 Restricted Control Cohort

The regression is estimated on the restricted control sample, retaining the 15 treated municipalities and the 24 control municipalities whose 2010 baseline social housing share (SH_SHARE_{2010}) and non-Western

immigrant share ($NW_IMMIG_SHARE_{2010}$) both exceed 5%. This restriction reduces the original set of 83 control municipalities to a more structurally comparable comparison group. The TWFE legislation coefficients in Table 2 are negative across all four specifications and for both outcomes, indicating a post-2018 narrowing tendency in treated municipalities relative to comparable controls. For the employment gap, $\beta = -0.695$ ($SE = 0.558, p = 0.214$) without controls and $\beta = -0.642$ ($SE = 0.541, p = 0.235$) with controls. Both estimates are statistically insignificant at the five percent level, with 95% confidence intervals that include zero. For the unemployment gap, $\beta = -0.674$ ($SE = 0.431, p = 0.118$) without controls and $\beta = -0.604$ ($SE = 0.473, p = 0.202$) with controls; these estimates are also statistically insignificant. Once the control group is restricted to structurally comparable municipalities, the legislation is associated with a gap-narrowing tendency rather than widening. However, the average treatment effect over the full 2018–2022 post-treatment window does not reach statistical significance in the compressed TWFE specification.

Table 2: TWFE Baseline DiD: Employment and Unemployment Gap, Restricted Sample

	(1a)	(1b)	(1c)	(1d)
Outcome	Emp. Gap	Emp. Gap	Unemp. Gap	Unemp. Gap
Controls	No	Yes	No	Yes
β (Treated $\times$ Post)	−0.695 (0.558)	−0.642 (0.541)	−0.674 (0.431)	−0.604 (0.473)
p-value	0.214	0.235	0.118	0.202
SH_SHARE	—	−0.009 (0.073)	—	−0.036 (0.063)
LOG_INCOME	—	+0.494 (9.269)	—	−6.545 (5.337)
NW_IMMIG_SHARE	—	+0.123 (0.321)	—	+0.150 (0.296)
R-squared	0.943	0.943	0.489	0.496
Observations	507	507	507	507
Treated municipalities	15	15	15	15
Control municipalities	24	24	24	24
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Sig. pre-trends, out of 7	0/7	0/7	2/7	2/7

Note: Cluster-robust standard errors in parentheses, clustered at the municipality level with 39 clusters. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, and $p < 0.10$. $Post_t = 1$ for $t \geq 2018$; reference year is 2017. The sample is restricted to municipalities with $SH_SHARE_{2010} > 5\%$ and $NW_IMMIG_SHARE_{2010} > 5\%$, yielding 15 treated municipalities, 24 control municipalities, and 507 municipality-year observations.

The event study results for the restricted sample, presented in fig 3 shows post-treatment coefficients are negative throughout 2018–2022 and grow monotonically in magnitude, consistent with a gradually accumulating legislation effect rather than an immediate structural break. The initial two post-treatment years, 2018 ($\delta = -0.503, p = 0.214$) and 2019 ($\delta = -0.641, p = 0.217$), are statistically insignificant, suggesting policy effect may not show up immediately after the law was introduced. From 2020 onwards, coefficients reach statistical significance: $\delta_{2020} = -1.358$ ($p = 0.046$), $\delta_{2021} = -1.606$ ($p = 0.042$), and $\delta_{2022} = -1.751$ ($p = 0.010$). By four years after implementation, the employment gap in treated muni-

papalities had narrowed by approximately 1.7–1.75% points relative to comparable control municipalities. On the other hand, unemployment gap exhibits a residual pre-trend within the restricted sample that prevents causal attribution, and its post-treatment dynamics are further confounded by the COVID-19 pandemic’s differential impact on NW immigrant unemployment in large urban municipalities.

5.3 COVID-Controlled Specification

From table 3 the employment gap legislation coefficient (β) is negative and statistically insignificant in both specifications: $\beta = -0.488$ ($SE = 0.521, p = 0.349$) without controls and $\beta = -0.438$ ($SE = 0.515, p = 0.395$) with controls. The COVID interaction (ϕ) is negative for the employment gap but statistically insignificant in both specifications: $\phi = -0.517$ ($SE = 0.343, p = 0.131$) without controls and $\phi = -0.513$ ($SE = 0.349, p = 0.141$) with controls. This finding indicates that the 2020–2021 pandemic did not produce a statistically detectable differential employment gap shock in treated municipalities relative to the restricted control cohort. The insignificance of ϕ for the employment gap is consistent with the descriptive evidence in Section 5.4 that the COVID-19 counter-cyclical employment gap narrowing was a sector-wide employment change across Denmark that happened broadly in both treated and comparable control municipalities, not mainly in the municipalities affected by the Parallel Society legislation. The event study in fig 4 satisfies parallel trends for the employment gap across all seven

Table 3: COVID-Controlled TWFE DiD: Employment and Unemployment Gap, Restricted Sample

	(1a)	(1b)	(1c)	(1d)
Outcome	Emp. Gap	Emp. Gap	Unemp. Gap	Unemp. Gap
Controls	No	Yes	No	Yes
β (Treated $\times$ Post)	−0.488 (0.521)	−0.438 (0.515)	−0.983 (0.511)	−0.903 (0.552)
p-value	0.349	0.395	0.055	0.102
ϕ (Treated $\times$ COVID)	−0.517 (0.343)	−0.513 (0.349)	+0.771** (0.277)	+0.754** (0.285)
p-value	0.131	0.141	0.005**	0.008**
R-squared	0.944	0.944	0.494	0.501
Observations	507	507	507	507
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Sig. pre-trends, excl. COVID	0/7	0/7	2/7	2/7

Note: Cluster-robust standard errors are reported in parentheses and clustered at the municipality level with 39 clusters.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, and $p < 0.10$. $\text{Post}_t = 1$ for $t \geq 2018$; $\text{COVID}_t = 1$ for $t \in \{2020, 2021\}$. In Equation 4, es_{2020} and es_{2021} are replaced by a single Treated $\times$ COVID dummy, ϕ , to avoid perfect collinearity. Standard controls are SH_SHARE, LOG_INCOME, and NW_IMMIG_SHARE. The restricted sample includes 15 treated and 24 control municipalities.

non-COVID pre-treatment years, validating causal identification. Post-treatment coefficients are negative throughout 2018–2022 and accumulate monotonically. The initial years, 2018 ($\delta = -0.503, p = 0.213$) and 2019 ($\delta = -0.641, p = 0.217$), are statistically insignificant, consistent with a two-year implementation lag.

The COVID interaction ($\phi = -1.482, SE = 0.709, p = 0.037$) captures the differential 2020–2021 shock in treated municipalities relative to control municipalities, net of national pandemic trends absorbed by year fixed effects. The 2022 coefficient constitutes the preferred post-pandemic point estimate: $\delta_{2022} = -1.751$ ($SE = 0.682, p = 0.010$) without controls and -1.700 ($SE = 0.653, p = 0.009$) with controls. Both estimates are significant at the 1% level, indicating that the employment gap in treated municipalities narrowed by approximately 1.70–1.75 percentage points relative to comparable controls by four years post-enactment.

The COVID interaction is positive and highly significant for the unemployment gap across both specifications: $\phi = +0.771$ ($SE = 0.277, p = 0.005$) without controls and $\phi = +0.754$ ($SE = 0.285, p = 0.008$) with controls. This indicates that the unemployment gap widened by approximately 0.75–0.77 percentage points more in treated municipalities than in comparable control municipalities during 2020–2021. This result is robust to the inclusion of controls. This suggests that COVID affected employment and unemployment in different ways. For employment, the effect appears linked to whether people remained attached to work. For unemployment, the effect likely reflects increased competition among job seekers, especially in treated urban municipalities with more contact-intensive labor markets.

5.4 Heterogeneous treatment effects model

The employment gap effect which is driven by newly designated municipalities by $\beta_1 = +1.814^*$ is the only employment gap coefficient significant at the conventional 5% level, while the continuously designated coefficient ($\beta_2 = +1.130$) is only weakly significant at the 10 % level. So the employment effect appears to be stronger where the policy arrived more suddenly. This pattern is contrary to a simple dose-response hypothesis and is consistent with abrupt network disruption in municipalities newly encountering dispersal obligations, generating sharper short-run employment gap widening than in areas where dispersal had been ongoing since 2010.

The unemployment gap effect is concentrated in continuously designated municipalities: $\beta_2 = +0.965^*$ is significant at the 5% level, while $\beta_1 = -0.354$ for newly designated municipalities is statistically insignificant. This suggests that long-term designated municipalities, especially large cities such as Copenhagen, Aarhus, and Odense, have deeper structural unemployment issues affecting non-Western immigrants. In newly designated municipalities, the policy may affect employment, but not necessarily registered unemployment. Some people may leave work or leave the labour force without appearing as unemployed.

Lastly, the falsification test largely supports causal attribution to the 2018 legislation. The exit-group coefficient is statistically insignificant for both outcomes in the preferred controlled specifications: $\beta_3 = +1.170$ ($p = 0.287$) for the employment gap and $\beta_3 = +0.947$ ($p = 0.103$) for the unemployment gap. That supports the argument that the main effects are more likely related to the 2018 legislation, not just pre-existing differences between municipalities.

Table 4: Heterogeneous Treatment Effects DiD: Employment and Unemployment Gap by Designation History

	(A)	(B)	(C)	(D)
Outcome	Emp. Gap	Emp. Gap	Unemp. Gap	Unemp. Gap
Controls	No	Yes	No	Yes
β_1 ($\text{New}_m \times \text{Post}_t$) [newly treated, $n = 4$]	+1.600*	+1.814*	−0.438	−0.354
	(0.803)	(0.878)	(0.607)	(0.676)
p-value	0.046*	0.039*	0.471	0.600
β_2 ($\text{Cont}_m \times \text{Post}_t$) [continuously treated, $n = 11$]	+1.062	+1.130	+0.928*	+0.965*
	(0.622)	(0.685)	(0.452)	(0.466)
p-value	0.088	0.099	0.040*	0.039*
β_3 ($\text{Exit}_m \times \text{Post}_t$) [exited, falsification, $n = 6$]	+1.220	+1.170	+1.096*	+0.947
	(1.128)	(1.098)	(0.529)	(0.581)
p-value [falsification passes if $p \geq 0.10$]	0.279	0.287	0.039*	0.103
R-squared	0.808	0.810	0.431	0.435
Observations	1,274	1,274	1,274	1,274
Municipality FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
$ \beta_1 $ vs. $ \beta_2 $	New > Cont	New > Cont	Cont > New	Cont > New
Falsification verdict	Passes	Passes	Fails	Passes

Note: Cluster-robust standard errors are reported in parentheses and clustered at the municipality level with 98 clusters.

*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, and $p < 0.10$. $\text{Post}_t = 1$ for $t \geq 2018$. New_m denotes municipalities designated in 2018 only: Helsingør, Guldborgsund, Vejle, and Silkeborg. Cont_m denotes municipalities designated in both 2010 and 2018: Copenhagen, Høje-Taastrup, Holbæk, Slagelse, Odense, Sønderborg, Esbjerg, Horsens, Kolding, Aarhus, and Viborg. Exit_m denotes municipalities designated in 2010 only and is used as the falsification group: Greve, Fredericia, Svendborg, Thisted, Aalborg, and Ishøj. Never-treated municipalities ($n = 77$) form the omitted reference group. Standard controls include SH_SHARE, LOG_INCOME, and NW_IMMIG_SHARE. The falsification test is considered to pass when the exited-group coefficient is statistically insignificant at conventional levels.

5.5 Callaway–Sant’Anna Heterogeneity-Robust Estimator

The Callaway–Sant’Anna doubly-robust estimator provides two contributions to the empirical strategy. First, for the employment gap, the CS estimator strengthens the TWFE event study findings: pre-treatment ATT² coefficients are clean with 1/7 significant year (attributing to the 2015 refugee crisis shock), and the post-treatment ATT accumulates monotonically to -1.964 pp by 2022 (SE = 0.851, $p = 0.021$). The CS point estimate is modestly larger in magnitude than the TWFE estimate (-1.964 vs. -1.751 pp), consistent with the more demanding conditional parallel trends requirement of the CS framework removing residual composition bias. Second, for the unemployment gap, the CS estimator partially resolves but does not eliminate the pre-trend concern: covariate conditioning reduces pre-trend significance from 3/7 to 1/7 significant pre years, but one residual pre-treatment year (2016: ATT = $+2.260$, $p = 0.017$) persists, and post-treatment ATT estimates collapse to insignificance under the covariate-adjusted specification. Taken together, these findings strengthen the paper’s primary conclusion that the 2018 Parallel Society legislation is associated with a gradual and statistically robust narrowing of the employment gap in treated municipalities by approximately 1.75–1.96 % points by 2022, while the unemployment gap evidence remains insufficient to support causal attribution.

6 Limitations

The study has several limitations to acknowledge before drawing policy conclusions. First, the analysis is conducted at the municipality level rather than at the level of the specific social housing estates targeted by the 2018 legislation. Since designated areas constitute only a fraction of each municipality’s residential and labour-market geography, employment effects concentrated within the affected estates may be diluted when aggregated to the municipality-year level. The estimated coefficients should therefore be interpreted as lower-bound estimates of the true effect operating within designated areas.

Second, treatment is coded as a time-invariant binary indicator, assigning equal treatment status to all 15 designated municipalities regardless of treatment intensity. This means that municipalities with several designated estates and large affected populations, such as Copenhagen, receive the same treatment weight as municipalities with only one smaller affected estate. The estimated β therefore represents an average treatment effect that may understate the impact in heavily affected municipalities and overstate it in lightly affected ones.

Third, while the employment gap satisfies the parallel trends assumption in the restricted sample, with zero of seven significant pre-treatment coefficients, the unemployment gap exhibits a persistent pre-trend, with two of seven significant pre-treatment coefficients and mean $|\delta| = 1.73$ % points. Consequently, unemployment-gap estimates are reported as descriptive upper bounds rather than causal estimates, and conclusions regarding unemployment should be interpreted with caution.

Fourth, spillover contamination may affect the treated-control comparison. The dispersal mechanism of the legislation may have relocated non-Western immigrant households from treated municipalities into untreated municipalities, including some municipalities in the restricted control cohort. If such relocation affects non-Western employment rates in control municipalities, the stable unit treatment value assumption may be violated. The preferred employment-gap estimates of approximately -1.70 to -1.75 percentage points should therefore be interpreted as reduced-form effects that may include spillover components.

Fifth, the post-treatment window is short. The panel extends only to 2022, while the legislation’s long-term objectives extend toward 2030. Demolition, rehousing, and redevelopment processes remain ongoing in many treated areas, meaning that the analysis cannot distinguish between temporary transition costs and

²Average Treatment Effect on the Treated after the policy starts

persistent long-run effects on non-Western immigrant employment outcomes.

Finally, the number of treated municipalities is small. With only 15 treated municipalities, and 24 municipalities in the restricted sample, cluster-robust inference has limited statistical power. Economically meaningful coefficients that do not reach conventional significance thresholds should therefore not be interpreted as evidence of no effect.

7 Conclusion

This paper examines the causal effect of Denmark’s 2018 Parallel Society Legislation on the native and non-Western immigrant employment and unemployment gaps across Danish municipalities using a difference-in-differences design applied to administrative panel data from Statistics Denmark for 2010–2022[12]. The analysis proceeds through a sequence of increasingly rigorous specifications: baseline TWFE, restricted control cohort, COVID-controlled augmented model, heterogeneous treatment effects, and Callaway–Sant’Anna robustness check.

The central finding is that the 2018 Parallel Society legislation is associated with a gradual and statistically robust narrowing of the native–non-Western immigrant employment gap in designated municipalities, with effects accumulating progressively over the post-treatment period. Within the preferred restricted control cohort, municipalities sharing comparable social housing concentrations and non-Western immigrant populations at the 2010 baseline, the employment gap in treated municipalities narrowed by approximately 1.70–1.75 % points relative to comparable controls by 2022: $\delta_{2022} = -1.751$, $SE = 0.682$, and $p = 0.010$. This estimate is significant at the 1 % level and is supported by the Callaway–Sant’Anna estimate of -1.964 % points ($p = 0.021$). The effect is not contemporaneous with the legislation’s enactment, as coefficients for 2018 and 2019 are statistically insignificant, but builds with a two-year implementation lag before reaching statistical significance from 2020 onwards. This temporal pattern is consistent with the dispersal and network-disruption mechanisms through which the legislation operates: disruption of social capital networks accumulated by non-Western immigrant translates into measurable employment consequences with a delay of approximately two years following forced relocation.

The heterogeneous treatment effects analysis reveals that this employment-gap narrowing is driven primarily by newly designated municipalities, where the legislation’s first-time designation in 2018 produced an abrupt institutional shock to established non-Western immigrant communities. This is reflected in $\beta_1 = +1.814$ ($p = 0.039$) for employment-gap widening in the full sample. The unemployment gap, by contrast, is concentrated in continuously designated municipalities, where long-term designation sustains structural non-Western immigrant unemployment concentrations: $\beta_2 = +0.965$ ($p = 0.039$). This asymmetric dose-response pattern across the two outcomes suggests that the legislation operates through qualitatively different channels depending on designation history: abrupt first-time designation disrupts employment networks, while persistent long-term designation maintains structural unemployment dynamics.

The COVID-19 pandemic introduced a significant differential shock to the unemployment gap in treated municipalities during 2020–2021: $\phi = +0.771$, $SE = 0.277$, and $p = 0.005$. This is consistent with the displacement of native workers from contact-intensive sectors into job search in the large urban labour markets of designated municipalities. This pandemic effect is separated from the legislation effect in the COVID-controlled specifications, ensuring that the preferred 2022 employment-gap estimate is not driven by pandemic dynamics.

Several important caveats qualify these findings. First, the analysis is conducted at the municipality level, which substantially attenuates effects concentrated within small designated estates and likely understates the true magnitude of the legislation’s impact on directly affected non-Western immigrant households.

Second, treatment is coded as a binary indicator, obscuring heterogeneity in treatment intensity across the 15 designated municipalities. Third, the unemployment gap cannot be given a causal interpretation due to persistent pre-trend violations that survive both covariate conditioning and the Callaway–Sant’Anna estimator. Fourth, the post-treatment window of five years is insufficient to assess whether the observed short-run employment-gap narrowing represents a transitional disruption cost that reverses as communities rebuild, or a persistent structural change. Finally, spillover contamination from dispersed households relocating to control municipalities likely biases the employment-gap coefficient upward. The evidence suggests that the 2018 legislation reduced the number of officially designated areas as documented by the OECD (2026)[7], but it also created short-run labour-market disruption for the non-Western immigrant population affected by the policy. It remains unclear whether these short-term costs will later be outweighed by long-term integration benefits. Future research using longer time periods and individual-level data will be needed to assess this.

8 REFERENCES

References

- [1] M. Bertrand, E. Duflo, and S. Mullainathan. How much should we trust differences-in-differences estimates? 119(1):249–275, 2004. doi: 10.1162/003355304772839588.
- [2] B. Callaway and P. H. C. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of Econometrics*, 225(2):200–230, 2021. doi: 10.1016/j.jeconom.2020.12.001.
- [3] A. P. Damm and M. Rosholm. Employment effects of spatial dispersal of refugees. CReAM Discussion Paper 05/06, Centre for Research and Analysis of Migration, University College London, 2005.
- [4] Landsbyggefonden. Beboere i den almene boligsektor 2025. Landsbyggefonden, 2025. URL <https://lbf.dk/viden/statistikker/beboerstatistik/beboere-i-den-almene-boligsektor-2025>. Accessed 27 April 2026.
- [5] M. Lundsteen. Displacing the other to unite the nation: The parallel society legislation in denmark. *European Urban and Regional Studies*, 30(3):261–281, 2023. doi: 10.1177/09697764231165202.
- [6] R. S. Nielsen, L. W. Nordberg, and H. T. Andersen. Taking the social out of social housing? recent developments, current tendencies, and future challenges to the danish social housing model. *Tidsskrift for boligforskning*, 6(2):136–151, 2023. doi: 10.18261/tfb.6.2.6.
- [7] OECD. *OECD Economic Surveys: Denmark 2026*, volume 2026 of *OECD Economic Surveys*. OECD Publishing, Paris, Jan. 2026. ISBN 978-92-64-32097-0. doi: 10.1787/3d6cb4b8-en. PDF ISBN: 978-92-64-77641-8; HTML ISBN: 978-92-64-88467-0.
- [8] Regeringen. Regeringens strategi mod ghettosering. Technical report, Ministeriet for Flygtninge, Indvandrere og Integration, København, May 2004. Electronic ISBN: 87-91320-86-0.
- [9] Regeringen. Ghettoen tilbage til samfundet: Et opgør med parallelsamfund i Danmark, Oct. 2010.
- [10] Regeringen. Ét Danmark uden parallelsamfund: Ingen ghettoer i 2030, 2018. URL <https://www.sm.dk/media/24960/et-danmark-uden-parallelsamfund-pdf.pdf>.

- [11] J. Sørensen, M. T. Sørensen, and F. K. Christensen. Nuances of “ghetto” policies in danish spatial planning. *Nordic Journal of Urban Studies*, 4(1):1–9, 2024. doi: 10.18261/njus.4.1.5.
- [12] Statistics Denmark. Statistics Denmark, 2026. URL <https://www.dst.dk/en>.
- [13] Transport-, Bygnings- og Boligministeriet. Liste over ghettoområder pr. 1. december 2018, 2018. URL <https://www.justiceinitiative.org/uploads/2e5883c5-99a4-4c14-b645-53dc4e0b3a21/tenants-danish-ministry-appendices-05272020.pdf>. J.nr. 2018-6411. Reproduced in appendix bundle hosted by Open Society Justice Initiative.

Appendix

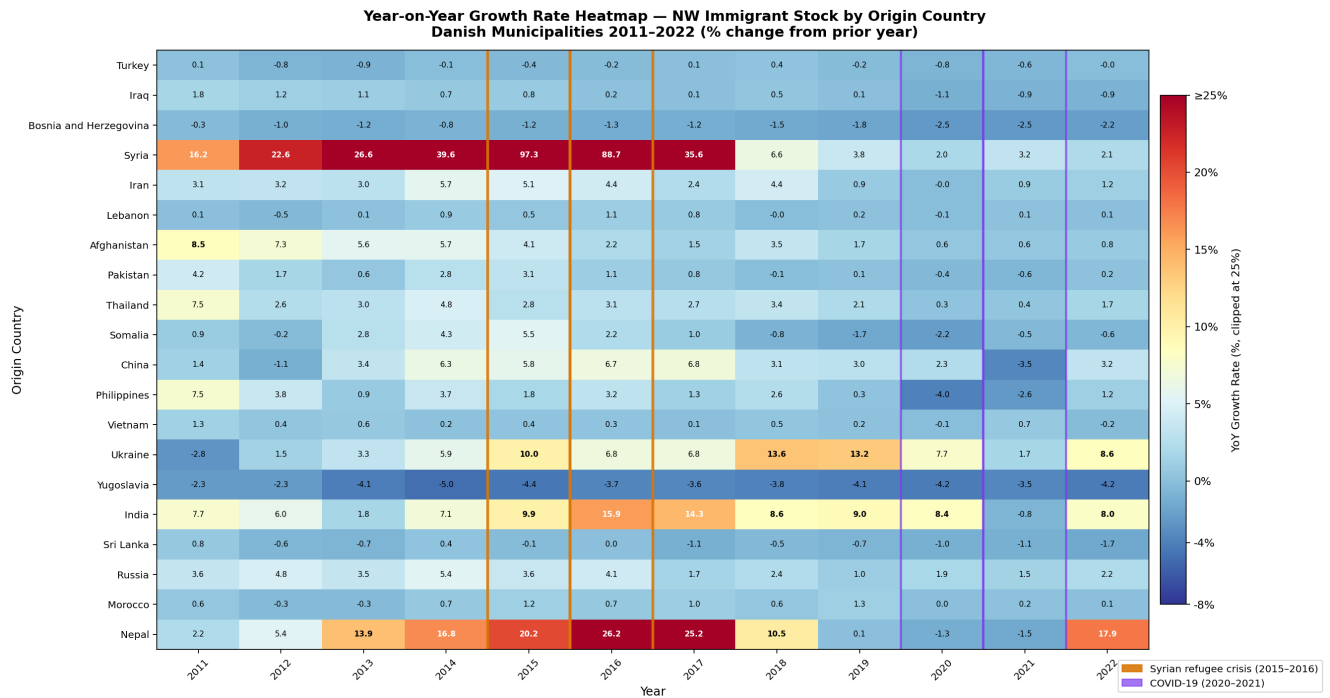


Figure 1: Year on Year Growth Rate of NW immigrant Stock by Origin Country from 2011-2022(% change from prior year)

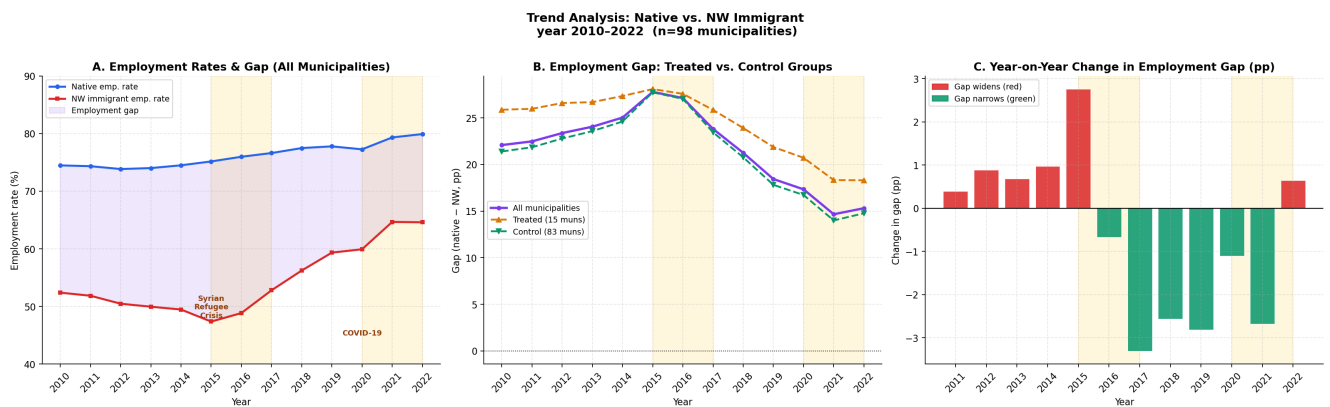


Figure 2: Time trend of the employment gap

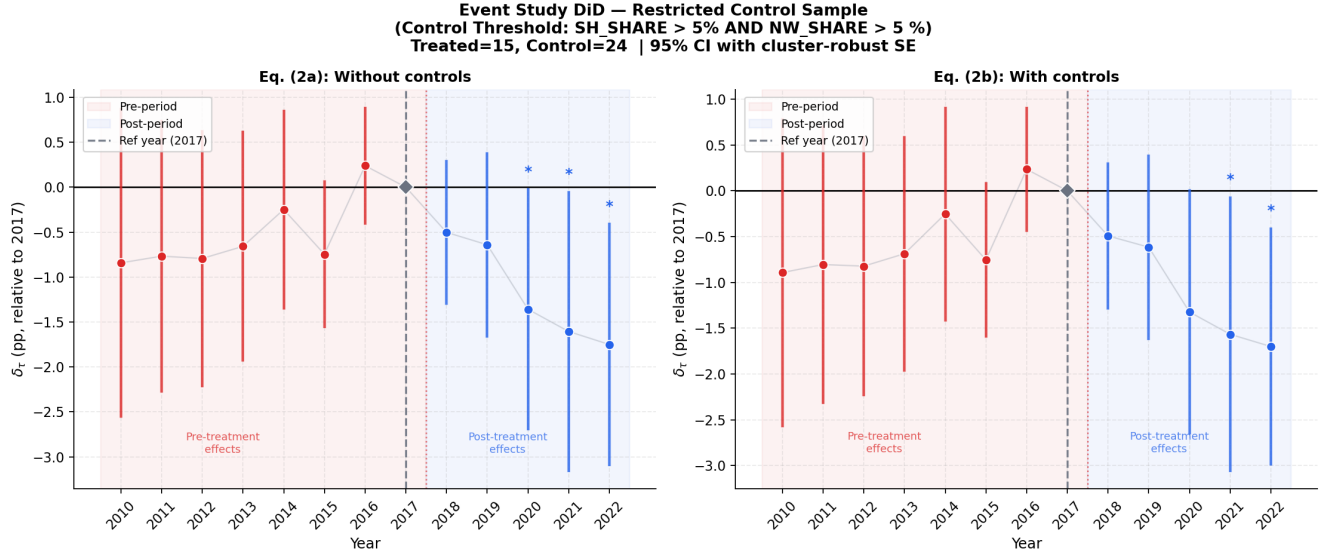


Figure 3: Event Study DiD — Restricted Control Cohort Employment Gap

Note: Event Study DiD, Restricted sample: treated = 15, control = 83. Red dots indicate pre-treatment coefficients; blue dots indicate post-treatment coefficients. Confidence intervals are cluster-robust 95% intervals. The significant pre-treatment coefficients in 2010–2011 and the large refugee-crisis disruption in 2015–2016 confirm that parallel trends do not hold for the full sample, motivating the restricted control cohort as the preferred specification. Asterisks (*) denote significance at the 5% level.

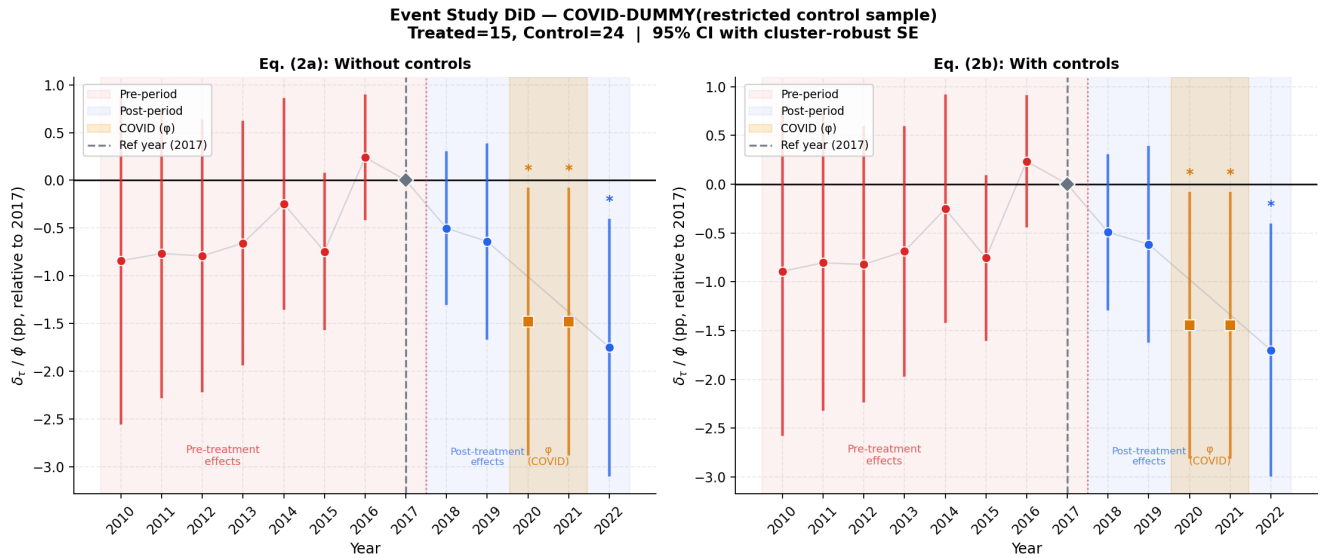


Figure 4: Event Study DiD — COVID-Controlled Specification, Restricted Control Sample

Note: Event Study DiD, COVID-controlled specification, restricted control sample: treated = 15, control = 24. Blue dots indicate post-treatment legislation coefficients (δ_t) for 2018, 2019, and 2022. Orange squares indicate the COVID interaction coefficient (ϕ), applied uniformly to 2020 and 2021. Red dots indicate pre-treatment coefficients. The grey diamond indicates the reference year, 2017. Confidence intervals are cluster-robust 95% intervals